Calculus II Lesson 23

1) Power Series: $\sum_{n=0}^{\infty} 2x^n = 2x^0 + 2x^1 + 2x^2 + \ldots = 2 + 2x^1 + 2x^2 + \ldots$

$\sum_{n=0}^{\infty} 2x^n$ is a Geometric Power Series with $a = 2$ and $r = x$

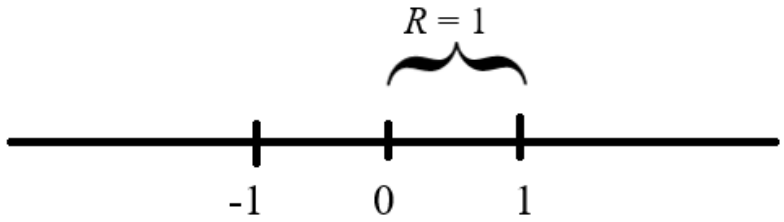
$\sum_{n=0}^{\infty} 2x^n$ converges to $\frac{a}{1-r} = \frac{2}{1-x}$ if $-1 < r < 1$ or $-1 < x < 1$

Note: $\sum_{n=0}^{\infty} 2x^n$ converges if x is inside the interval $(-1, 1)$;

$\sum_{n=0}^{\infty} 2x^n$ diverges if is not in the interval $(-1, 1)$

Interval of Convergence is $(-1, 1)$, centered at $x = 0$.

Radius of Convergence is $R = 1$.



a) When $x = -0.9$, $\sum_{n=0}^{\infty} 2x^n = \frac{2}{1-x} =$ ____?____

b) When $x = -0.5$, $\sum_{n=0}^{\infty} 2x^n = \frac{2}{1-x} =$ ____?____

c) When $x = 0$, $\sum_{n=0}^{\infty} 2x^n = \frac{2}{1-x} =$ ____?____

d) When $x = 0.6$, $\sum_{n=0}^{\infty} 2x^n = \frac{2}{1-x} =$ ____?____

e) When $x = -1.5$, $\sum_{n=0}^{\infty} 2x^n =$ ____?____

Note: $x = -1.5$ is NOT inside the interval $(-1, 1)$.

f) When $x = 2$, $\sum_{n=0}^{\infty} 2x^n =$ ____?____ (converges or diverges?)

Note: $x = 2$ is NOT inside the interval $(-1, 1)$.

g) Complete the table of values for the rational function $f(x) = \frac{2}{1-x}$.

x	$f(x) = \frac{2}{1-x}$
-0.9	
-0.5	
0	
0.6	
-1.5	
2	

2) Power Series: $\sum_{n=0}^{\infty} 2^{-n} \cdot (-1)^{n+1} (x-2)^n$

$$\sum_{n=0}^{\infty} 2 \cdot (-1)^{n+1} (x-2)^n = 2 \cdot (-1)^1 (x-2)^0 + 2 \cdot (-1)^2 (x-2)^1 + 2 \cdot (-1)^3 (x-2)^2 + \ldots$$

$$= -2 + 2(x-2)^1 - 2(x-2)^2 + 2(x-2)^3 - 2(x-2)^4 + \ldots$$

Note: $\sum_{n=0}^{\infty} 2 \cdot (-1)^{n+1} (x-2)^n$ is a Geometric Series with $a = -2$ and $r = x-2$

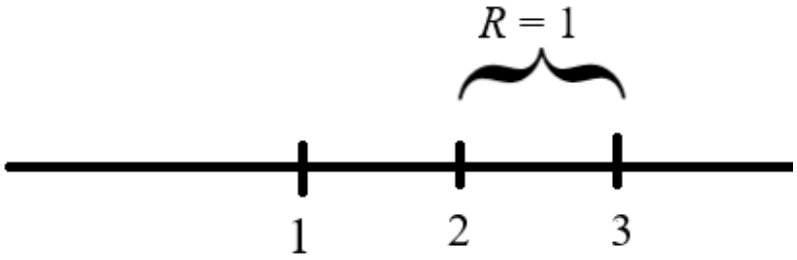
Hence $\sum_{n=0}^{\infty} 2 \cdot (-1)^{n+1} (x-2)^n$ converges to $\frac{a}{1-r} = \frac{-2}{1+(x-2)}$ if $-1 < x-2 < 1$ or $1 < x < 3$

Note: $\sum_{n=0}^{\infty} 2 \cdot (-1)^{n+1} (x-2)^n$ converges if x is inside the interval $(1, 3)$;

$\sum_{n=0}^{\infty} 2 \cdot (-1)^{n+1} (x-2)^n$ diverges if is not in the interval $(1, 3)$

Interval of Convergence is $(1, 3)$, centered at $x = 2$.

Radius of Convergence is $R = 1$.



a) When $x = 1.5$, $\sum_{n=0}^{\infty} 2\cdot(-1)^{n+1}(x-2)^n = \frac{-2}{1+(x-2)} =$ ___?___ Note: $x = 1.5$ is inside the interval $(1, 3)$.

b) When $x = 2$, $\sum_{n=0}^{\infty} 2\cdot(-1)^{n+1}(x-2)^n = \frac{-2}{1+(x-2)} =$ ___?___ Note: $x = 2$ is inside the interval $(1, 3)$.

c) When $x = 2.5$, $\sum_{n=0}^{\infty} 2\cdot(-1)^{n+1}(x-2)^n = \frac{-2}{1+(x-2)} =$ ___?___ Note: $x = 2.5$ is inside the interval $(1, 3)$.

d) When $x = 2.9$, $\sum_{n=0}^{\infty} 2\cdot(-1)^{n+1}(x-2)^n = \frac{-2}{1+(x-2)} =$ ___?___ Note: $x = 2.9$ is inside the interval $(1, 3)$.

e) When $x = 3.5$, $\sum_{n=0}^{\infty} 2\cdot(-1)^{n+1}(x-2)^n =$ ___?___ (converges or diverges?) Note: $x = 3.5$ is NOT inside the interval $(1, 3)$.

f) When $x = 0.5$, $\sum_{n=0}^{\infty} 2\cdot(-1)^{n+1}(x-2)^n =$ ___?___ (converges or diverges?) Note: $x = 0.5$ is NOT inside the interval $(1, 3)$.

g) Complete the table of values for the rational function $f(x) = \frac{2}{1-x}$.

x	$f(x) = \frac{2}{1-x}$
1.5	
2	
2.5	
2.9	
3.5	
0.5	

3) Power Series: $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}=\frac{(-1)^1(x-3)^0}{2^0}+\frac{(-1)^2(x-3)^1}{2^1}+\frac{(-1)^3(x-3)^2}{2^2}+\frac{(-1)^4(x-3)^3}{2^3}+\ldots$

$$=\text{-}1+\frac{(x-3)}{2}-\frac{(x-3)^2}{4}+\frac{(x-3)^3}{8}-\ldots$$

$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ is a Geometric Power Series with $a=-1$ and $r=\frac{1}{2}(x-3)$.

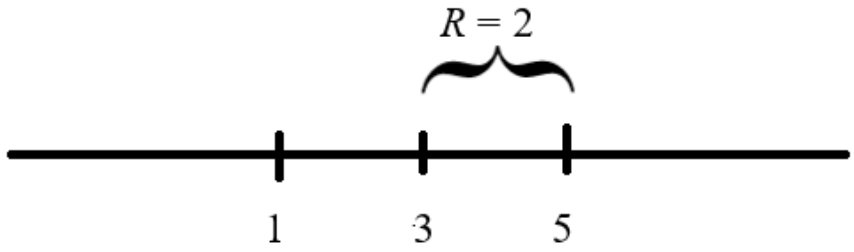
$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ converges to $\frac{a}{1-r}=\frac{-1}{1--\frac{1}{2}(x-3)}=\frac{-1}{1+\frac{1}{2}(x-3)}$ if $-1<-\frac{1}{2}(x-3)<1$ or $1<x<5$

Note: $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ converges if x is inside the interval $(1, 5)$;

$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ diverges if is not in the interval $(1, 5)$

Interval of Convergence is $(1, 5)$, centered at $x=3$.

Radius of Convergence is $R=2$.



3) Power Series: $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}=\frac{(-1)^1(x-3)^0}{2^0}+\frac{(-1)^2(x-3)^1}{2^1}+\frac{(-1)^3(x-3)^2}{2^2}+\frac{(-1)^4(x-3)^3}{2^3}+\ldots$

$$=\text{-}1+\frac{(x-3)}{2}-\frac{(x-3)^2}{4}+\frac{(x-3)^3}{8}-\ldots$$

$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ is a Geometric Power Series with $a=-1$ and $r=\frac{1}{2}(x-3)$.

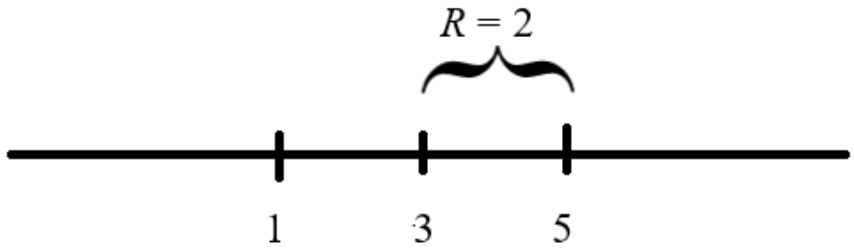
$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ converges to $\frac{a}{1-r}=\frac{-1}{1--\frac{1}{2}(x-3)}=\frac{-1}{1+\frac{1}{2}(x-3)}$ if $-1<-\frac{1}{2}(x-3)<1$ or $1<x<5$

Note: $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ converges if x is inside the interval $(1, 5)$;

$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{2^n}$ diverges if is not in the interval $(1, 5)$

Interval of Convergence is $(1, 5)$, centered at $x=3$.

Radius of Convergence is $R=2$.



4) Power Series: $\sum_{n=0}^{\infty}\frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}} = \frac{2(-1)^1(x-4)^0}{3^1}+\frac{2(-1)^2(x-4)^1}{3^2}+\frac{2(-1)^3(x-4)^2}{3^3}+\frac{2(-1)^4(x-4)^3}{3^3}+\ldots$

$$= -\frac{2}{3}+\frac{2(x-4)}{3^2}-\frac{2(x-4)^2}{3^3}+\frac{2(x-4)^3}{3^3}+\ldots$$

$\sum_{n=0}^{\infty}\frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}}$ is a Geometric Power Series with $a=-\frac{2}{3}$ and $r=-\frac{1}{3}(x-4)$.

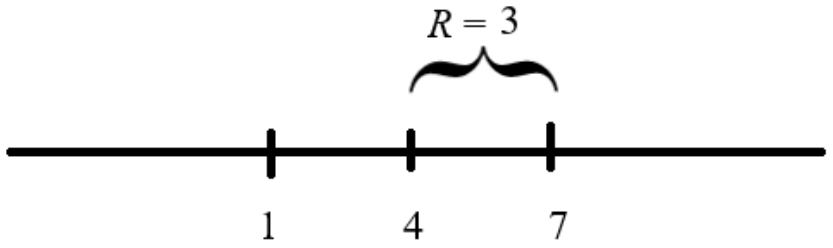
$\sum_{n=0}^{\infty}\frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}}$ converges to $\frac{a}{1-r}=\frac{-2/3}{1--\frac{1}{3}(x-4)}=\frac{-2/3}{1+\frac{1}{3}(x-4)}$ if $-1<-\frac{1}{3}(x-4)<1$ or $1<x<7$

Note: $\sum_{n=0}^{\infty}\frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}}$ converges if x is inside the interval $(1, 7)$;

$\sum_{n=0}^{\infty}\frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}}$ diverges if is not in the interval $(1, 7)$

Interval of Convergence is $(1, 7)$, centered at $x=4$.

Radius of Convergence is $R=3$.



a) When $x = 1.5$, $\displaystyle\sum_{n=0}^{\infty} \frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}} = \frac{-2/3}{1+\frac{1}{3}(x-4)} =$ ____?____ Note: $x = 1.5$ is inside the interval $(1, 7)$.

b) When $x = 2$, $\displaystyle\sum_{n=0}^{\infty} \frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}} = \frac{-2/3}{1+\frac{1}{3}(x-4)} =$ ____?____ Note: $x = 2$ is inside the interval $(1, 7)$.

c) When $x = 0.5$, $\displaystyle\sum_{n=0}^{\infty} \frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}} =$ ____?____ Note: $x = 0.5$ is NOT inside the interval $(1, 7)$.

d) When $x = 8$, $\displaystyle\sum_{n=0}^{\infty} \frac{2(-1)^{n+1}(x-4)^n}{3^{n+1}} =$ ____?____ Note: $x = 8$ is NOT inside the interval $(1, 7)$.

e) Complete the table of values for the rational function $f(x) = \dfrac{2}{1-x}$.

x	$f(x) = \dfrac{2}{1-x}$
1.5	
2	
0.5	
8	

5) Find a geometric power series for the function $f(x) = \dfrac{1}{4 - x}$ centered at 0.

Note: $\dfrac{1}{4 - x} = \dfrac{1/4}{(4 - x)/4} = \dfrac{1/4}{1 - x/4} = \dfrac{a}{1 - r}$

Divide each term by 4

a) a = ?

b) r = ?

c) Recall: $\sum_{n=0}^{\infty} a \cdot r^n = a \cdot r^0 + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots = a + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots$

The Geometric power series $\sum_{n=0}^{\infty} a \cdot r^n$ converges to $\dfrac{a}{1 - r}$ if $-1 < r < 1$.

$f(x) = \dfrac{1}{4 - x} = \dfrac{a}{1 - r} = \sum_{n=0}^{\infty} a \cdot r^n =$? if $-1 < r < 1$. (Multiple Choice)

A) $\sum_{n=0}^{\infty} 4\left(\dfrac{x}{4}\right)^n$ B) $\sum_{n=0}^{\infty} \dfrac{1}{4}\left(\dfrac{x}{4}\right)^n$ C) $\sum_{n=0}^{\infty} \dfrac{1}{4}(x)^n$ D) $\sum_{n=0}^{\infty} \left(\dfrac{1}{4}\right)^n$

Note: The answer in part (c) is called a power series representation of the function $f(x) = \dfrac{1}{4 - x}$ centered at $x = 0$.

6) Find a geometric power series for the function $f(x) = \frac{3}{4 - x}$ centered at 0.

Note: $\frac{3}{4 - x} = \frac{(3)/4}{(4 - x)/4} = \frac{3/4}{1 - x/4} = \frac{a}{1 - r}$

Divide each term by 4

a) $a =$?

b) $r =$?

c) Recall: $\sum_{n=0}^{\infty} a \cdot r^n = a \cdot r^0 + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots = a + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots$

The Geometric power series $\sum_{n=0}^{\infty} a \cdot r^n$ converges to $\frac{a}{1 - r}$ if $-1 < r < 1$.

$f(x) = \frac{3}{4 - x} = \frac{a}{1 - r} = \sum_{n=0}^{\infty} a \cdot r^n =$? if $-1 < r < 1$. (Multiple Choice)

A) $\sum_{n=0}^{\infty} 3\left(\frac{x}{4}\right)^n$ B) $\sum_{n=0}^{\infty} \frac{1}{4}\left(\frac{x}{4}\right)^n$ C) $\sum_{n=0}^{\infty} \frac{3}{4}\left(\frac{x}{4}\right)^n$ D) $\sum_{n=0}^{\infty} \left(\frac{4}{4}\right)^n$

Note: The answer in part (c) is called a power series representation of the function $f(x) = \frac{3}{4 - x}$ centered at $x = 0$.

7) Find a geometric power series for the function $f(x) = \dfrac{1}{1 - 3x}$ centered at $c = 0$.

$Note: \dfrac{1}{1 - 3x} = \dfrac{a}{1-r}$

a) $a =$?

b) $r =$?

c) Recall: $\sum_{n=0}^{\infty} a \cdot r^n = a \cdot r^0 + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots = a + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots$

The Geometric power series $\sum_{n=0}^{\infty} a \cdot r^n$ converges to $\dfrac{a}{1-r}$ if $-1 < r < 1$.

$f(x) = \dfrac{1}{3-x} = \dfrac{a}{1-r} = \sum_{n=0}^{\infty} a \cdot r^n =$? if $-1 < r < 1$. (Multiple Choice)

A) $\sum_{n=0}^{\infty} \frac{1}{2}(3x)^n$ B) $\sum_{n=0}^{\infty} (3x)^n$ C) $\sum_{n=0}^{\infty} \frac{1}{4}(3x)^n$ D) $\sum_{n=0}^{\infty} 2(3x)^n$

Note: The answer in part (c) is called a power series representation of the function $f(x) = \dfrac{1}{1-3x}$ centered at $x = 0$.

8) Find a geometric power series for the function $f(x) = \dfrac{1}{3 - x}$ centered at $x = 1$.

Note: To find a power series centered at $x = 1$, replace x with "$x - 1 + 1$" and simplify.

Hence, $\dfrac{1}{3-x} = \dfrac{1}{3-(x-1+1)} = \dfrac{1}{3-(x-1)-1} = \dfrac{1}{2-(x-1)} = \dfrac{1/2}{2/2-(x-1)/2} = \dfrac{1/2}{1-\frac{1}{2}(x-1)} = \dfrac{a}{1-r}$

Divide each term by 2

a) a = ? _____

b) r = ? _____

c) Recall: $\sum_{n=0}^{\infty} a \cdot r^n = a \cdot r^0 + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots = a + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots$

The Geometric power series $\sum_{n=0}^{\infty} a \cdot r^n$ converges to $\dfrac{a}{1-r}$ if $-1 < r < 1$.

$f(x) = \dfrac{1}{3-x} = \dfrac{a}{1-r} = \sum_{n=0}^{\infty} a \cdot r^n =$? if $-1 < r < 1$. (Multiple Choice)

A) $\sum_{n=0}^{\infty} \left(\dfrac{x-1}{2}\right)^n$ B) $\sum_{n=0}^{\infty} \dfrac{1}{2}(x-1)^n$ C) $\sum_{n=0}^{\infty} \dfrac{1}{4}\left(\dfrac{x-1}{4}\right)^n$ D) $\sum_{n=0}^{\infty} \dfrac{1}{2}\left(\dfrac{x-1}{2}\right)^n$

Note: The answer in part (c) is called a power series representation of the function $f(x) = \dfrac{1}{3-x}$ centered at $x = 1$.

9) Find a geometric power series for the function $f(x) = \dfrac{3}{3x + 4}$ centered at $c = 0$.

Note: $f(x) = \dfrac{3}{3x-4} = \dfrac{-3}{-3x+4} = \dfrac{-3}{4-3x} = \dfrac{-3/4}{4/4-3x/4} = \dfrac{-3/4}{1-\frac{3}{4}x} = \dfrac{a}{1-r}$

Divide each term by 4

a) a = ? _____

b) r = ? _____

c) Recall: $\sum_{n=0}^{\infty} a \cdot r^n = a \cdot r^0 + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots = a + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots$

The Geometric power series $\sum_{n=0}^{\infty} a \cdot r^n$ converges to $\dfrac{a}{1-r}$ if $-1 < r < 1$.

$f(x) = \dfrac{3}{3x-4} = \sum_{n=0}^{\infty} a \cdot r^n =$? if $-1 < r < 1$. (Multiple Choice)

A) $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right)\left[\frac{-3x}{4}\right]^n$ B) $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right)\left[\frac{-3x}{2}\right]^n$ C) $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)\left[\frac{-3x}{4}\right]^n$ D) $\sum_{n=0}^{\infty} \left(\frac{-3}{4}\right)\left[\frac{3x}{4}\right]^n$

Note: The answer in part (c) is called a power series representation of the function $f(x) = \dfrac{3}{3x+4}$ centered at $x = 0$.

10) Find a geometric power series for the function $f(x) = \dfrac{5}{2x - 3}$ centered at $c = -3$.

Note: $f(x) = \dfrac{5}{2x-3} = \dfrac{5}{2[x-(-3)+(-3)]-3} = \dfrac{5}{2(x-(-3))-6-3} = \dfrac{5}{2(x+3)-9}$

$= \dfrac{5}{-9+2(x+3)} = \dfrac{5/(-9)}{-9/(-9)+2(x+3)/(-9)} = \dfrac{-5/9}{1-\frac{2}{9}(x+3)} = \dfrac{a}{1-r}$

a) a = ? _____

b) r = ? _____

c) Recall: $\sum_{n=0}^{\infty} a \cdot r^n = a \cdot r^0 + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots = a + a \cdot r^1 + a \cdot r^2 + a \cdot r^3 + \ldots$

The Geometric power series $\sum_{n=0}^{\infty} a \cdot r^n$ converges to $\dfrac{a}{1-r}$ if $-1 < r < 1$.

Power series representation of the function $f(x) = \dfrac{5}{2x-3}$ is $\sum_{n=0}^{\infty} ar^n =$? (Multiple Choice)

A) $\sum_{n=0}^{\infty} \left(\frac{1}{9}\right)\left[\frac{2}{9}(x+3)\right]^n$

B) $\sum_{n=0}^{\infty} \left(\frac{-1}{9}\right)\left[\frac{2}{9}(x+3)\right]^n$

C) $\sum_{n=0}^{\infty} \left(\frac{-5}{9}\right)\left[\frac{2}{9}(x+3)\right]^n$

D) $\sum_{n=0}^{\infty} \left(\frac{-4}{9}\right)\left[\frac{4}{9}(x+3)\right]^n$

Note: The answer in part (c) is called a power series representation of the function $f(x) = \dfrac{5}{2x-3}$ centered at $x = -3$.